Spatiotemporal Analysis of Coastal Seagrass Dynamics in Upper Florida Keys

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Sentinel-2 Imagery of Upper Florida Keys.
https://commons.wikimedia.org/wiki/File:Florida_Keys_(from_Lower_Matecumbe_Key_to_Key_Largo)_by_Sentinel-2.jpg

Seagrass

- Submerged flowering plants that grow in shallow coastal waters. Seagrass provides essential habitats for diverse assemblages of marine life, stabilizes sediments, sequesters carbon, and improves overall water quality.

- Highly sensitive to changing environmental conditions, especially water clarity and light availability.

- Parameters of interest that are likely to affect seagrass health.
  - Turbidity
  - Water Clarity / light availability
  - Salinity
  - Temperature

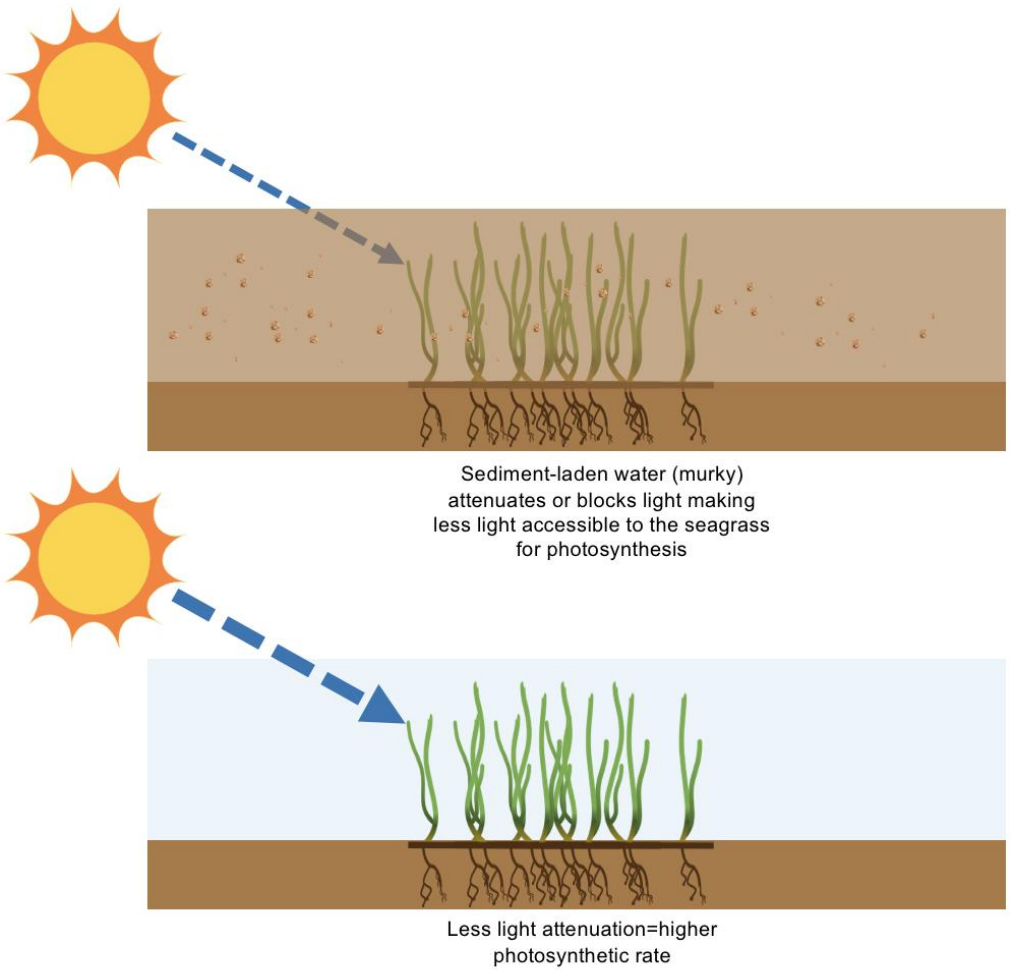
https://floridawildlifefederation.org/expanding-seagrass-beds-are-good-news/

Our Objectives

- Assess relationships between seagrass presence and water quality parameters (turbidity, salinity, temperature and light availability).
- Map spatial distribution patterns of seagrass and compare against coastal conditions.
- Analyze temporal trends using time series data to identify changes in seagrass dynamics over time.
- Identify key environmental drivers influencing observed spatial and temporal variability.

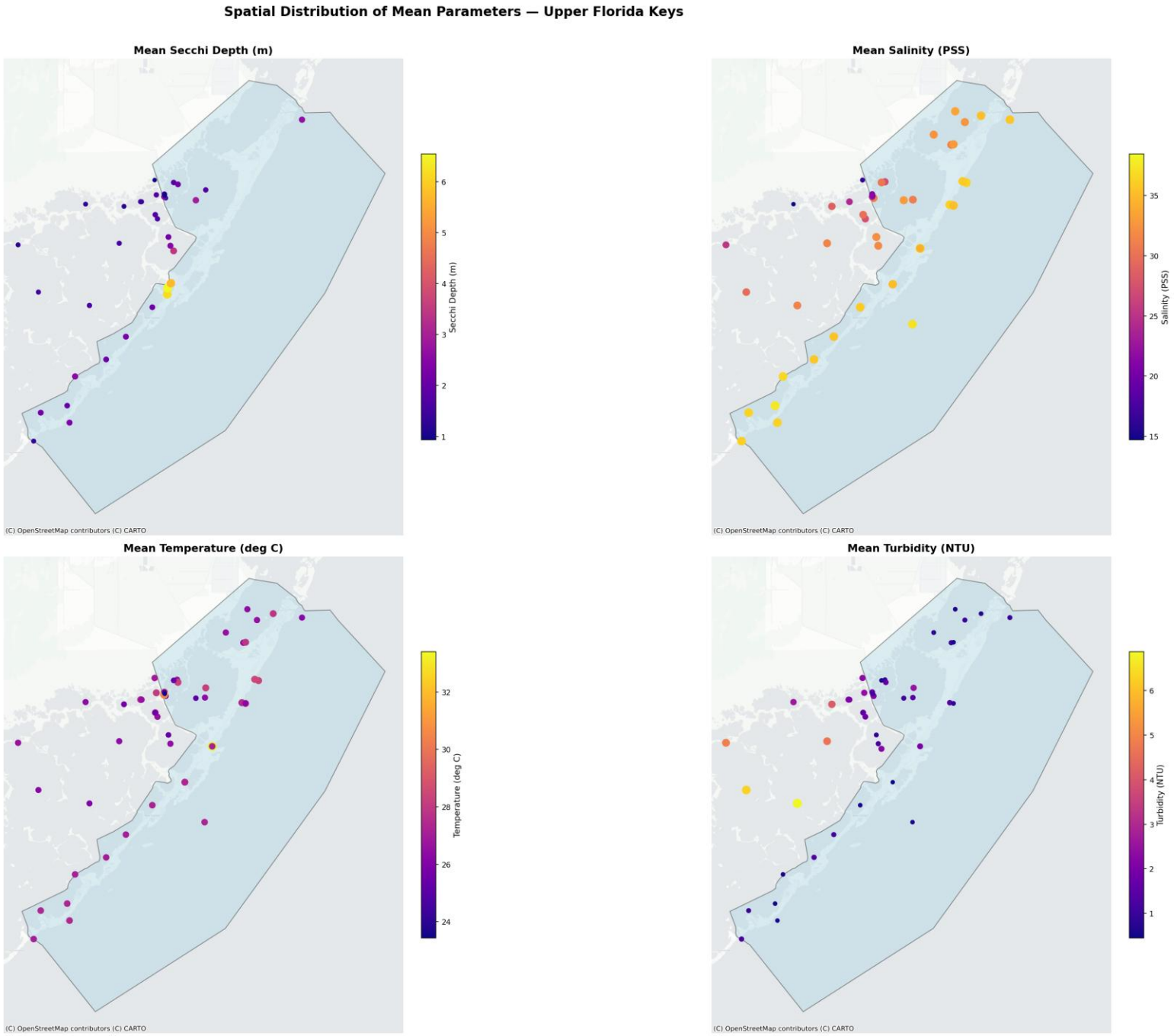
Water Quality

- Water quality refers to environmental conditions such as turbidity, temperature, and salinity that influence seagrass growth.
- Turbidity is the most important because it controls how much light reaches the seafloor. When turbidity increases, less light is available, which reduces photosynthesis. Temperature and salinity also affect seagrass by causing stress and limiting where it can survive.



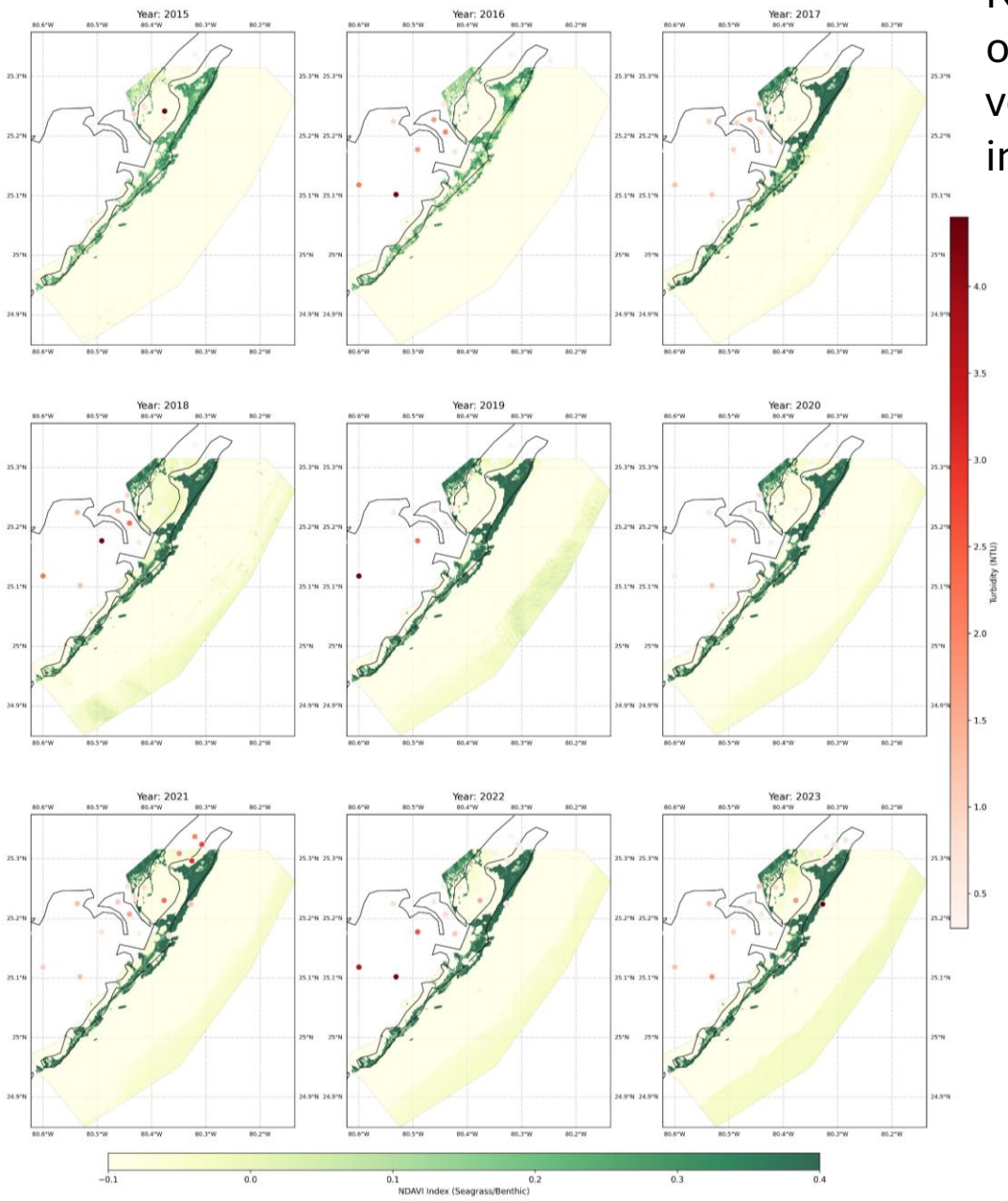
Spatial Distribution of mean parameters

- These maps show how water quality varies across the area. The colors represent the values, where lighter means higher and darker means lower. Overall mean turbidity in our study area is generally low, but we can still see that areas with higher turbidity tend to have lower water clarity, which affects seagrass growth.



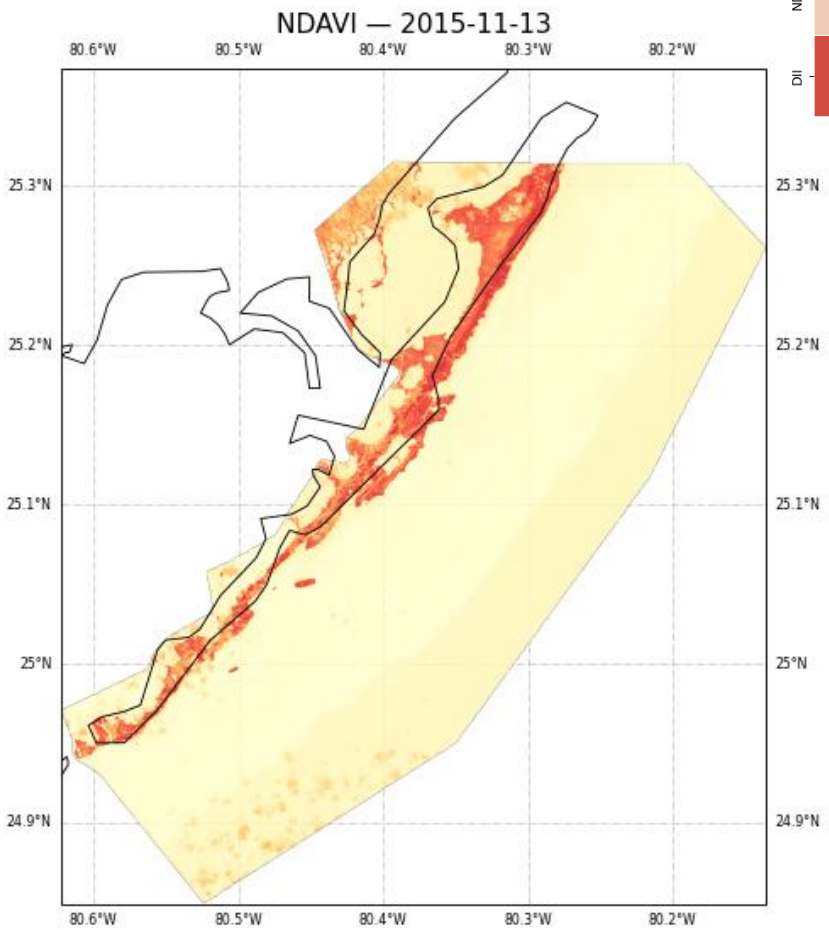
https://github.com/ssullivan9661/seagrass-analysis/blob/main/figures/spatial_means.png

Spatial Analysis

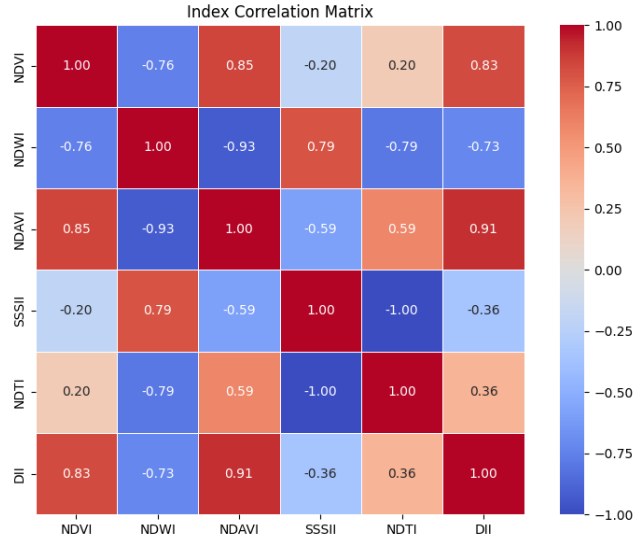


https://github.com/ssullivan9661/seagrass-analysis/blob/main/figures/Annual_Full_Coverage_Analysis.jpg

NDAVI is a direct indicator of observable seagrass vegetation verified with correlation to Depth invariant index (DII).



https://github.com/ssullivan9661/seagrass-analysis/blob/main/figures/ndti_animation_2015_2024.gif



https://github.com/ssullivan9661/seagrass-analysis/blob/main/figures/Index_Correlation_Matrix.png

Temporal Analysis

BLUE - NDVI (Normalized Difference Vegetation Index)

- Baseline. General vegetation health

ORANGE – NDAVI (Normalized Difference Aquatic Vegetation Index)

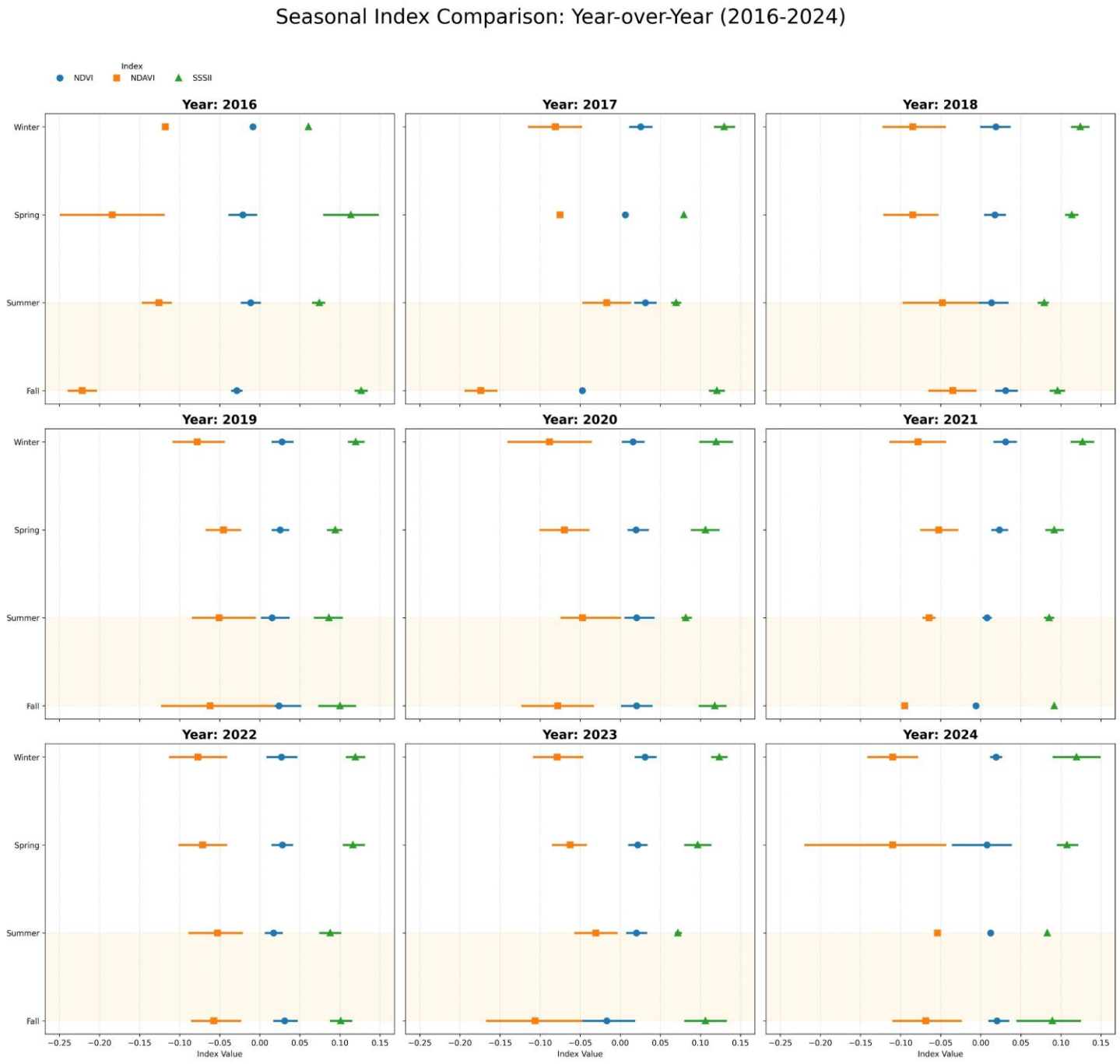
- Aquatic vegetation health. Better suited for submerged vegetation

GREEN – SSSII (Submerged Seagrass Spectral Index)

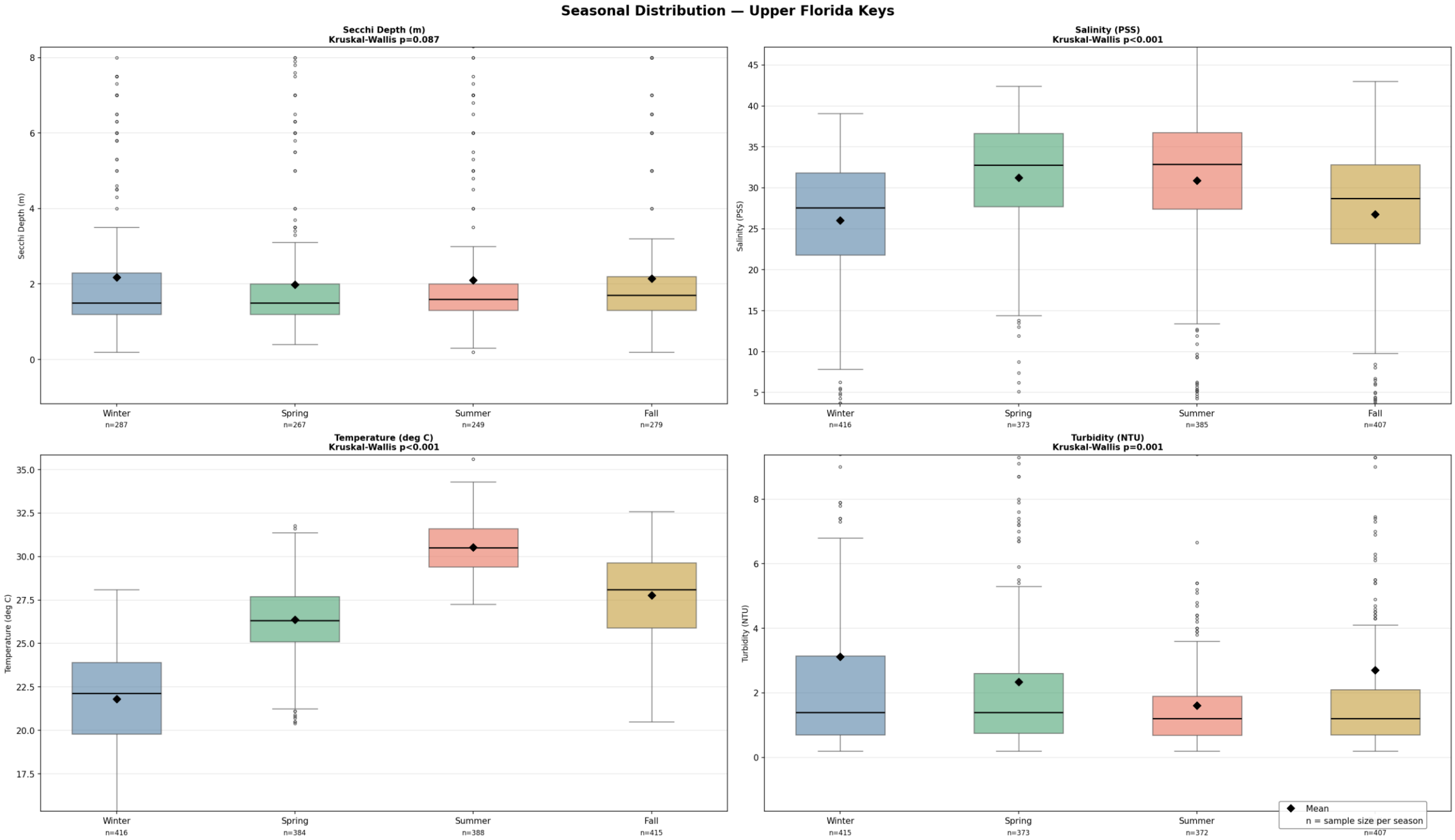
- Focuses on water column, close to 0 = visible seagrass / less debris in water

- Strong agreement in seasonal trends. Atypical data from 2016 suggests lingering fallout from 2015 die-off event (Kavanagh, 2016).

2017 major divergence to 2018 snap-back shows natural tendency for ecosystems in disaster recovery.



https://github.com/ssullivan9661/seagrass-analysis/blob/main/figures/indices_seasonal_comparison.jpg



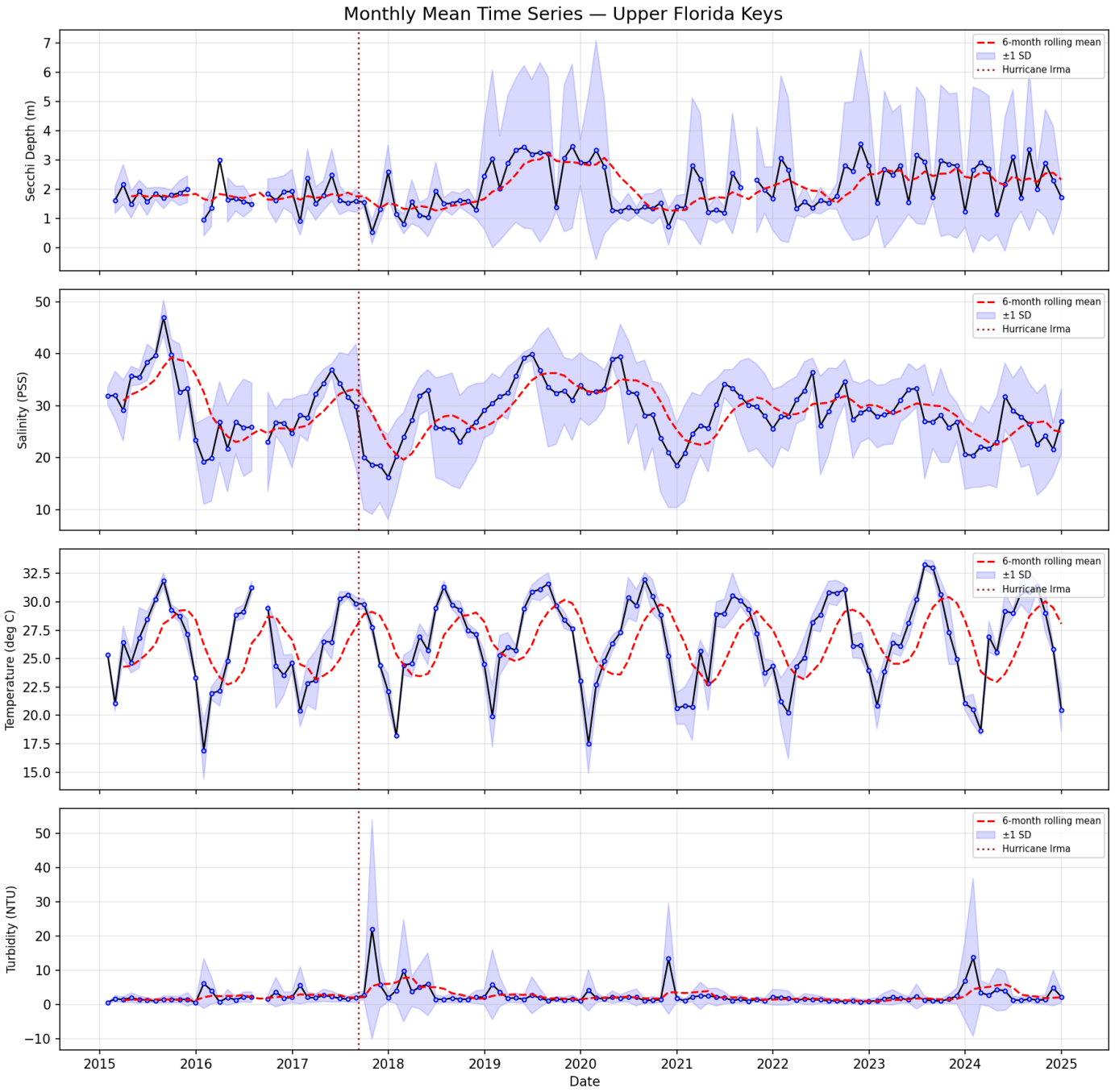
https://github.com/ssullivan9661/seagrass-analysis/blob/main/figures/seasonal_boxplots.png

Temporal Analysis cont.

Key Observations:

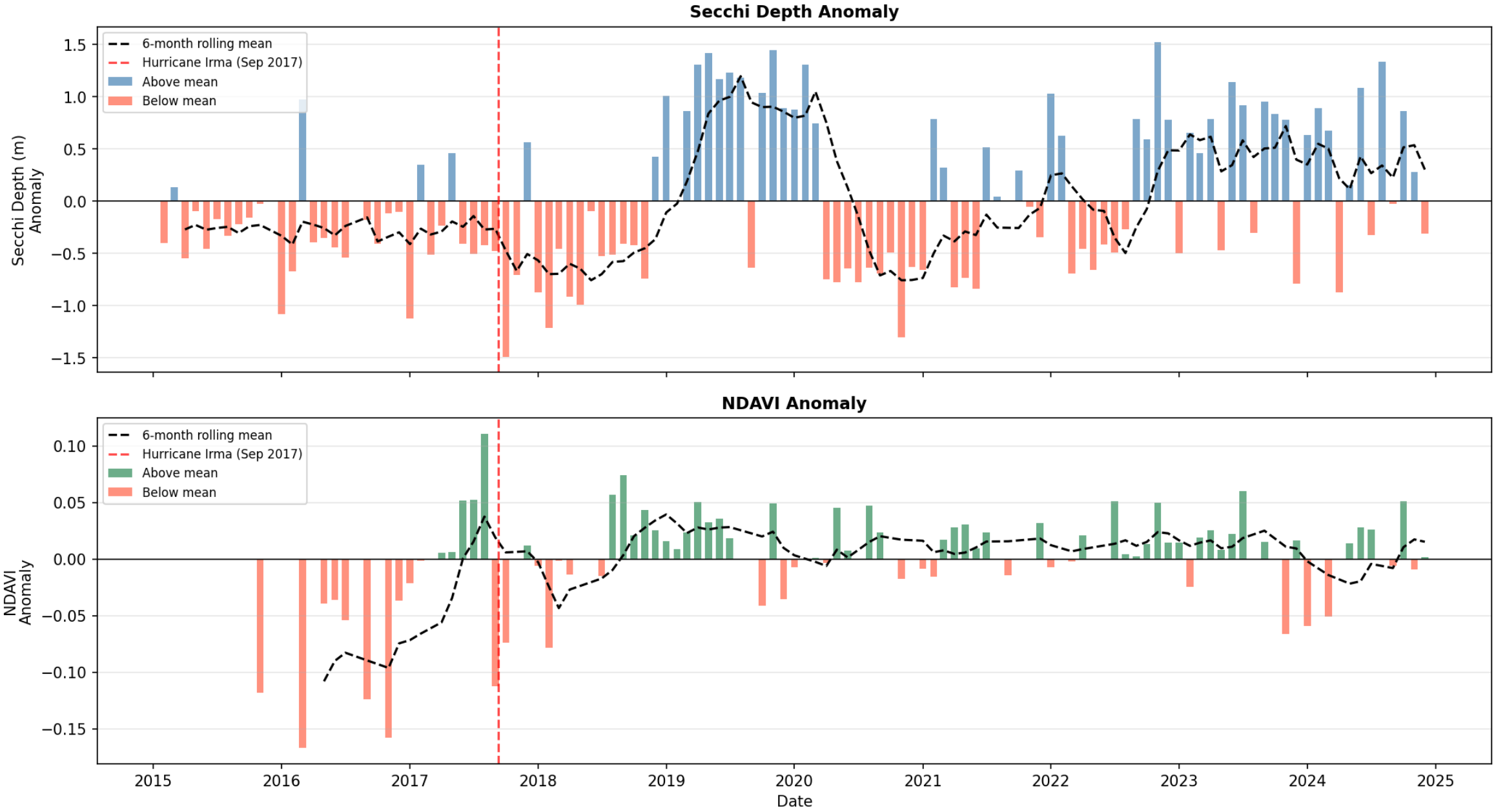
- Clear seasonal patterns for salinity and temperature while turbidity and water clarity (Secchi depth) are more event driven.
- Clear impact from Hurricane Irma (2017), turbidity spike, water clarity drop.
- Gradual stabilization and improved clarity following Irma.

Shaded region represents ±1 standard deviation, indicating variability in measurements.



https://github.com/ssullivan9661/seagrass-analysis/blob/main/figures/time_series.png

Anomaly Plot — Secchi Depth & NDAVI
Upper Florida Keys 2015–2024



https://github.com/ssullivan9661/seagrass-analysis/blob/main/figures/anomaly_plot.png

Conclusions

- Turbidity strongly influences seagrass distribution in an inverse relationship.
- Evidence the seagrass community of study is a strong “disaster-recovery” resilient ecosystem,
  - following 2015-drought caused die-off,
  - 2017 Hurricane Irma,
- Observed recovery aligns with documented seagrass expansion across Florida Bay (Hall et al., 2021).
- Overall, positive trend with small natural variations

References

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- Carlson, P. R., Yarbro, L. A., & Barber, T. R. (2017). *2015 Florida Bay seagrass die-off*. Florida Fish and Wildlife Conservation Commission. https://www.researchgate.net/publication/316554225_2015_Florida_Bay_Seagrass_Die-Off

- https://github.com/ssullivan9661/seagrass-analysis